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Practical Number: 3Practical Name: Use Case Diagram.Hardware Requirements - pentium 4 processor (2.4 GHz), 128 mb RAM, standard keyboard n mouse, coted monitor.Software Requirements - StatUML, Windows XP.Outcome - can produce the requirements in use case diagram.Objective - prepare a Requirement document in by using use case diagram.Theory -

According to the UML Specification a use case diagram is "a diagram that shows the relationships among actors & use cases within a system".

Use case diagrams are often used to

provide an overview of all or part of the usage requirements for a system organizations in the form of an essential model or a business Model.

Communicate the scope of a development project.

Model your analysis of your usage requirements in the form of a system use case model.

Use case models should be developed from the point of view of your project stakeholders and not from the (often technical) point of view developers. These

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Category guidelines for:

Use cases

Actors

Relationships

System Boundary Boxes

1) Use cases

- A use case describes a sequence of actions that provide a measurable value to an actor. A use case is drawn as a horizontal ellipse on a UML Use Case diagram.

1. Use case diagram Name begin with a Strong Verb.
2. Name use cases using domain terminology.
3. Place your primary user cases in the top-left corner of the diagram.
4. Imply timing considerations by stacking use cases.

2) Actors

- An actor is a person, organization, or external system that plays a role in one or more interaction with your system. Actors are typically drawn as stick figures on UML Use Case diagrams.

- 1) Place your primary actor(s) in the top-left corner of the diagram.
- 2) Draw actors to the outside of a use case diagram.
- 3) Name actors with singular, business-relevant nouns.
- 4) Associate each actor with one or more use cases.

- 5) Actors Model roles, not positions.
- 6) Use <<System>> to indicate System actors.
- 7) Actors don't Interact with one another.
- 8) Introduce an actor called "Time" to initiate Scheduled events.

3) Relationships.

There are several types of relationships that may appear on a use case diagram:-

An association between an actor and a use case

An association between two use cases

A generalization between two actors

A generalization between two use cases.

Associations are depicted as lines connecting two modeling elements with an optional open-headed arrowhead on one end of the line indicating the direction of the initial invocation of the relationship. Generalizations are depicted as a close-headed arrow with the arrow pointing towards the more general modeling element.

- 1) Indicate An Association between An Actor and A Use case if the actor appears within the Use case logic.
- 2) Avoid Arrowheads On Actor - Use case relationships.
- 3) Apply <<include>> when you know Exactly when to invoke the use case.

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- 4) Apply <<extend>> when a use case may be invoked across several use case steps.
- 5) Introduce <<extend>> associations sparingly.
- 6) Generalize use cases when a single condition results in significantly new business logic.
- 7) Do not apply <<uses>>, <<includes>>, or <<extends>>
- 8) Avoid more than two levels of use case associations.
- 9) Place an included use case to the right of the invoking use case.
- 10) Place an extending use case below the parent use case.
- 11) Apply the "is like" Rule to use case generalization.
- 12) Place an inheriting use case below the parent use case.
- 13) Apply the "is like" Rule to actor inheritance.
- 14) Place an inheriting actor below the parent actor.

4. System Boundary Boxes

The rectangle around the use cases is called the system boundary box and as the name suggests it indicates the scope of your system - the use cases inside the rectangle represent the functionality that you intend to implement.

- 1) Indicate Release Scope with a System Boundary Box.
- 2) Avoid meaningless system Boundary Boxes.

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Creating Use case diagrams

We start by identifying as many actors as possible. you should ask how the actors interact with the system to identify an initial set of use cases. Then, on the diagram, you connect the actors with the use cases with which they are involved. If actors supply information, initiates the use case, or receives any information as a result of the use case, then there should be an association between them.

Conclusion - The use case diagram was made successfully by following the steps

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